

MERIDIAN & ASSOCIATES LLP

Advisory | Assurance | Tax | Consulting

Case Study: Energy Sector

Streamlining Regulatory Compliance and Financial Reporting
for a Major Upstream Oil & Gas Producer

CLIENT CONFIDENTIAL - BLINDED

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Meridian & Associates LLP | New York | Chicago | San Francisco | London | Singapore

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1. Executive Summary

Meridian & Associates LLP was engaged by a Fortune 500 exploration and production (E&P) company to deliver an integrated suite of assurance, tax, and advisory services spanning a three-year engagement period. The client, one of the largest independent upstream oil and gas producers in North America, operates across multiple prolific basins including the Permian, Eagle Ford, Bakken, and DJ Basin, with approximately \$18 billion in annual revenue and over 8,500 employees.

The engagement encompassed an integrated financial statement audit, SOX Section 404 internal controls assessment, SEC reporting advisory, and a comprehensive tax provision automation initiative. Meridian deployed a cross-functional team of 47 professionals representing our Assurance, Tax, and Advisory service lines, working in close coordination with the client's finance, accounting, tax, and IT departments across six office locations.

This case study provides a detailed overview of the engagement challenges, Meridian's phased approach, specialist team composition, and the quantifiable business outcomes delivered to the client over the course of the relationship.

Engagement at a Glance

Metric	Detail
Client Revenue	\$18.2B (FY 2025)
Industry	Oil & Gas - Upstream E&P
Engagement Duration	36 months (2023-2026)
Team Size	47 professionals (peak)
Service Lines	Audit, Tax, Advisory
Total Engagement Value	\$6.8M (cumulative)
Client Satisfaction	4.8 / 5.0 (annual survey)

2. Client Background

The client is a publicly traded independent E&P company headquartered in Houston, Texas, with operations spanning the continental United States and select international assets in Canada and Colombia. The company's asset portfolio includes approximately 2.1 million net acres across four major basins, with proved reserves of 1.8 billion barrels of oil equivalent (BOE). Daily production averages approximately 485,000 BOE/day, split roughly 62% oil and condensate, 23% natural gas, and 15% natural gas liquids (NGLs).

The company completed two significant acquisitions in the 18 months preceding the engagement - a \$3.2 billion bolt-on acquisition in the Permian Basin and a \$1.1 billion entry into the DJ Basin - which added complexity to the financial reporting environment. Additionally, the company had recently transitioned its accounting methodology from the full-cost method to the successful-efforts method, a change that required extensive retrospective adjustments and ongoing judgment-intensive accounting determinations.

3. Key Challenges

3.1 Commodity Price Volatility and Financial Reporting

The client's financial results are highly sensitive to commodity price fluctuations. WTI crude oil prices ranged from \$62 to \$93 per barrel during the engagement period, while Henry Hub natural gas prices varied between \$1.80 and \$5.40 per MMBtu. This volatility created significant challenges in several areas:

- Impairment testing of long-lived assets under ASC 360, requiring quarterly assessment of proved and unproved properties across four basins with differing economic profiles
- Derivative financial instrument accounting under ASC 815, with the client maintaining a hedging program covering approximately 65% of near-term production using a mix of swaps, collars, and three-way collars
- Revenue recognition under ASC 606 for complex marketing and transportation arrangements, including take-or-pay contracts, index-based pricing with quality differentials, and intercompany elimination of midstream revenues
- Going-concern and liquidity assessments during periods of depressed commodity prices, particularly given the company's \$4.5 billion revolving credit facility with borrowing base redetermination provisions

3.2 Asset Retirement Obligation (ARO) Estimation

The client operates approximately 12,400 producing wells across its portfolio, each requiring individual ARO estimation under ASC 410-20. The complexity of ARO estimation was compounded by several factors:

- Multi-jurisdictional regulatory requirements spanning Texas, New Mexico, North Dakota, Colorado, Wyoming, and two Canadian provinces, each with distinct plugging and abandonment standards
- Wide variation in estimated settlement costs ranging from \$35,000 per well for shallow vertical wells to \$850,000 for deep horizontal wells with multi-stage completions
- Discount rate determination challenges in a rising interest rate environment, requiring regular reassessment of credit-adjusted risk-free rates
- Legacy liabilities from the two recent acquisitions, where historical ARO records were incomplete or based on outdated cost assumptions

3.3 Full-Cost to Successful-Efforts Transition

The client's recent transition from the full-cost method to the successful-efforts method of accounting for oil and gas properties under ASC 932 introduced pervasive complexity. Under the full-cost method, all exploration costs had been capitalized in a single cost center. The transition required the client to retrospectively identify and expense exploration costs for dry holes and geological/geophysical expenditures that did not result in proved reserves, spanning a 15-year historical period encompassing over 4,200 individual well files.

This transition also required recalculation of depletion under the unit-of-production method at a property-by-property level rather than the full-cost pool level, necessitating detailed reserve allocation and cost assignment for each of the client's approximately 340 distinct producing properties.

4. Meridian's Approach

Meridian structured the engagement as a three-phase program, with overlapping workstreams to maximize efficiency and minimize disruption to the client's operations. Our approach leveraged deep industry expertise, purpose-built analytical tools, and a collaborative delivery model that embedded our professionals alongside the client's teams.

4.1 Phase 1: Assessment & Stabilization (Months 1-6)

The initial phase focused on gaining a comprehensive understanding of the client's financial reporting environment, identifying control deficiencies, and establishing a remediation roadmap.

- **Financial Reporting Diagnostic:** Conducted a 6-week diagnostic assessment of the client's close process, identifying 23 bottlenecks that contributed to a 22-business-day close cycle. Key findings included manual journal entry processes for intercompany eliminations, lack of automated reconciliation for revenue accruals, and inconsistent application of the successful-efforts method across basins.
- **SOX 404 Scoping & Risk Assessment:** Performed top-down risk assessment identifying 14 significant accounts and 8 relevant assertions. Identified 187 key controls across 12 business processes, with 34 controls requiring remediation or redesign. Highest-risk areas included production revenue, derivative accounting, and property cost capitalization.
- **Tax Provision Gap Analysis:** Evaluated the client's tax provision process, which relied on a combination of spreadsheets, legacy Corptax modules, and manual calculations. Identified \$2.8 million in potential tax overpayments related to incorrectly classified intangible drilling costs (IDCs) and suboptimal utilization of percentage depletion allowances.
- **Technology Assessment:** Mapped the client's financial reporting technology stack, including SAP S/4HANA (core ERP), Quorum (production accounting), Allegro (commodity trading/risk management), and BlackLine (account reconciliation). Identified integration gaps and data quality issues that contributed to manual workarounds.

4.2 Phase 2: Execution & Remediation (Months 7-24)

The execution phase represented the core of the engagement, with concurrent workstreams addressing audit, controls remediation, and tax automation.

- **Integrated Audit Execution:** Delivered integrated financial statement audits for FY 2023 and FY 2024, with a combined team of 28 assurance professionals including 4 partners and 6 managers. Deployed our proprietary MeridianAI analytics platform to analyze 100% of journal entries (approximately 1.2 million annually) for anomaly detection, replacing the prior auditor's sample-based approach.
- **SOX Remediation Program:** Worked with the client to remediate 34 identified control deficiencies, including 6 material weaknesses and 11 significant deficiencies. Designed and implemented automated controls for revenue recognition, lease accounting, and intercompany transactions. Conducted walkthroughs and operating effectiveness testing for all 187 key controls across four quarterly testing cycles.
- **ARO Revaluation Initiative:** Assembled a specialist team including petroleum engineers and environmental

specialists to revalue ARO obligations for all 12,400 wells. Developed a proprietary statistical model that incorporated basin-specific cost data, regulatory requirements, and historical plugging and abandonment costs from state databases. The revaluation resulted in a net adjustment of \$142 million to the ARO liability, with corresponding adjustments to asset carrying values and accretion expense.

- **Tax Provision Automation:** Implemented OneSource Tax Provision integrated with SAP S/4HANA, replacing the legacy spreadsheet-based process. Configured the system for 42 tax jurisdictions (federal, 38 states, 3 Canadian provinces), including automated calculation of percentage depletion, IDC deductions, and enhanced oil recovery credits. Developed automated book-to-tax difference tracking for over 200 temporary differences.

4.3 Phase 3: Optimization & Transition (Months 25-36)

The final phase focused on sustaining improvements, optimizing processes, and transitioning knowledge to the client's internal teams.

- **Close Cycle Optimization:** Implemented a structured close calendar with automated task tracking, eliminating 14 of the 23 identified bottlenecks. Introduced parallel processing for basin-level close activities and automated the consolidation of intercompany transactions across 28 legal entities.
- **SEC Reporting Enhancement:** Redesigned the client's SEC reporting process for Forms 10-K, 10-Q, and 8-K, including implementation of Workiva (formerly Wdesk) for collaborative document preparation and XBRL tagging. Reduced the SEC filing preparation timeline from 18 days to 11 days.
- **Knowledge Transfer & Training:** Conducted 32 hours of structured training sessions covering successful-efforts accounting, derivative instrument accounting, and ARO estimation methodology. Delivered comprehensive process documentation including 47 updated accounting policies and 23 procedure manuals.
- **Continuous Monitoring Framework:** Established a continuous monitoring program for key financial reporting controls, leveraging automated exception reporting and dashboard-based oversight. Transitioned monitoring responsibility to the client's internal audit team with ongoing advisory support.

5. Team Composition

Meridian deployed a cross-functional team drawn from three service lines, with deep energy sector expertise at every level. The team was structured to provide continuous coverage across the client's six office locations while maintaining centralized quality oversight. The senior team drew on Meridian's deepest bench of industry specialists, led by our Energy & Utilities Practice Lead Patricia Hoffman.

Role	Name	Service Line	Specialization	Allocation
Lead Engagement Partner	Patricia Hoffman, CPA	Energy & Utilities	E&P Audit / SOX	40%
Advisory Partner	James O'Sullivan, CPA	Manufacturing & SCM	Operations / Controls	25%
Sr. Manager - Tax	Rachel Okonkwo, CPA, MS	Tax	Tax Provision	70%
Sr. Manager - SALT	Sophia Vasquez, CPA, JD	Tax	SALT / Multi-State	50%
Sr. Manager - Technology	David Kim, PMP, SAP	Advisory	ERP / SAP S/4HANA	60%
Manager - Risk	Maria Santos, CPA, CIA	Advisory	Internal Controls	80%
Sr. Consultant - SAP	Marcus Wright	Consulting	SAP FICO	100%
Sr. Consultant - SAP	Kwame Asante	Consulting	SAP S/4HANA FICO	100%
Consultant - Data Eng.	Jessica Huang	Consulting	Data Engineering	90%
Staff - PMO	Derek Williams	Advisory	PMO Coordination	100%

In addition to the senior team listed above, the engagement was supported by 12 senior associates, 18 associates, and 7 staff-level professionals. Total engagement hours over the 36-month period were approximately 68,000, with peak staffing of 47 professionals during year-end audit execution periods.

Team Qualifications

- 100% of partners and senior managers hold active CPA licenses
- Average of 16 years of energy sector experience among partners and managers
- Patricia Hoffman brings CEM (Certified Energy Manager) credentials and Six Sigma Black Belt certification with deep upstream E&P expertise
- James O'Sullivan provides operational excellence perspective from 26 years in manufacturing and supply chain, with direct relevance to field operations optimization
- Rachel Okonkwo and Sophia Vasquez bring complementary tax expertise spanning federal provision and multi-state SALT, critical for the client's 42-jurisdiction tax footprint
- David Kim's 11 full-lifecycle SAP S/4HANA implementations ensured seamless ERP integration for the tax provision automation workstream
- Marcus Wright and Kwame Asante provided dedicated SAP FICO configuration support, with Kwame holding SAP S/4HANA certification
- Jessica Huang built the data pipelines connecting Quorum, Allegro, and SAP for the MeridianAI analytics platform

6. Quantifiable Outcomes

The engagement delivered measurable, sustained improvements across the client's financial reporting, compliance, and tax functions. The following outcomes were validated through independent measurement and client confirmation.

6.1 Financial Close & Reporting

Metric	Before Meridian	After Meridian	Improvement
Close Cycle (business days)	22 days	14 days	35% reduction
SEC Filing Prep Time	18 days	11 days	39% reduction
Manual Journal Entries	2,340 / quarter	890 / quarter	62% reduction
Restatement Risk Items	14 identified	0 remaining	100% resolved
Audit Adjustments (Year 1 vs 3)	23 proposed	4 proposed	83% reduction

6.2 SOX 404 Compliance

Metric	Before Meridian	After Meridian	Improvement
Material Weaknesses	6	0	100% remediated
Significant Deficiencies	11	1	91% remediated
Control Deficiencies	34 total	3 remaining	91% resolved
Automated Controls (%)	28%	67%	+39 pct points
Testing Exceptions	42 / cycle	7 / cycle	83% reduction

6.3 Tax & Cost Savings

Category	Amount	Description
R&D Tax Credits Identified	\$4.2M	Previously unclaimed credits for drilling technology innovation
IDC Reclassification Benefit	\$2.8M	Corrected classification of intangible drilling costs
Percentage Depletion Recovery	\$1.4M	Optimized depletion method elections across properties
Tax Provision Efficiency	\$380K/yr	Annual cost savings from automated tax provision process
Audit Fee Reduction (Year 3)	\$420K	Efficiency gains shared with client through reduced fees

7. Engagement Timeline

Phase	Period	Key Milestones
Phase 1	Jan - Jun 2023	Diagnostic complete, SOX scoping, tax gap analysis
Audit Y1	Jul - Dec 2023	FY 2023 integrated audit, initial SOX testing
Phase 2a	Jan - Jun 2024	SOX remediation, tax automation implementation
Audit Y2	Jul - Dec 2024	FY 2024 audit, ARO revaluation, controls retesting
Phase 2b	Jan - Jun 2025	SEC reporting redesign, close optimization
Audit Y3	Jul - Dec 2025	FY 2025 audit, clean SOX opinion achieved
Phase 3	Jan - Mar 2026	Knowledge transfer, continuous monitoring handoff

8. Client Testimonial

"Meridian's team brought a level of energy-sector expertise that we had not experienced with our prior auditors. Their ability to integrate audit, tax, and advisory services into a cohesive program saved us significant time and resources. The identification of \$4.2 million in R&D credits alone justified the engagement economics, and the improvement in our close cycle has fundamentally changed how our finance organization operates."

- Chief Financial Officer, [Client Name Redacted]

9. Lessons Learned & Transferable Insights

- **Early Integration Pays Dividends:** Co-locating audit, tax, and advisory teams from Day 1 reduced information asymmetry and eliminated an estimated 2,400 hours of duplicative data requests over the engagement.
- **Technology-Enabled Audit Is Transformative:** Deploying MeridianAI for 100% journal entry testing replaced statistical sampling and identified 14 anomalous transactions in Year 1 that the prior sample-based approach had missed.
- **Specialist Resources Drive Value:** The inclusion of petroleum engineers and environmental specialists on the ARO workstream produced a \$142M revaluation that a generalist team would not have identified.
- **Automation Requires Change Management:** The tax provision automation initiative required more change management effort than initially scoped. We added dedicated training resources in Phase 2, increasing the original training budget by 40% to ensure sustainable adoption.